

4(a)(i)	<i>any two from:</i> microtubules / spindle (fibres), are attached to, centromere / kinetochore ; (spindle) fibres, contract / shorten / retract ; A microtubules disassemble / AW move / pull, (sister) chromatids / (daughter) chromosomes, to opposite poles ;	2
4(a)(ii)	deoxyribose <u>and</u> phosphate <u>and</u> base ; <i>further detail</i> <i>one from</i> diagram / description, of phosphate group and base attached to deoxyribose ; one correctly named base or A,T C and G ;	2
4(b)(i)	<i>any two from:</i> allow DNA replication to continue ; <i>idea that</i> stem cell to, self renew / keep dividing ; <i>idea that</i> stem cells differentiate into (named) specialised cells ; <i>idea that</i> mitosis produces cells for, cell replacement / tissue repair ; (maintains telomeres, so) prevents the loss of genes ;	2

Question	Answer	Marks
4(b)(ii)	<i>any one from:</i> (high level of telomerase activity) allows ,continuous / uncontrolled, mitosis / cell division ; idea of allowing shorter cell cycle ;	1
4(b)(iii)	<i>any three from:</i> DNA binds to (complementary section of) mRNA / hydrogen bonds form between the complementary bases (of DNA and RNA) ; mRNA cannot bind to the ribosome ; ribosome cannot, move along the mRNA ; consequence e.g. no sites for tRNA to bind / tRNA cannot bring attached amino acid ; tRNA anticodons cannot form hydrogen bonds with mRNA codons ; AVP ; e.g. <i>idea that</i> DNA enters nucleus to bind and mRNA then too large to leave via nuclear pores	3

Question	Answer	Marks
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